VIPP Report

DeepInfer Prostate MRI Segmentation Evaluation

PROMISE12 Test Data (30 Cases)

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Report

Introduction

In this report, I evaluated the `prostate-segmenter` model from the Docker image `deepinfer/prostate` for prostate MRI segmentation on a 30-case PROMISE12 test set stored in the local `test_data` folder. The original data were provided as MetaImage `.mhd/.raw` files and were converted to `.nrrd` format for inference. Before evaluation, the converted `.nrrd` files were verified against the original `.mhd` files so that image size, voxel spacing, origin, orientation, and voxel values matched correctly. This source-versus-converted verification passed for all **60/60** converted files (30 MRI volumes and 30 segmentation masks).

A description of the model can be found at: https://www.deepinfer.org/models/prostate-segmenter/

The dataset description can be found at: https://promise12.grand-challenge.org/Details/

The evaluation was carried out as a whole-gland segmentation task using the binary prostate masks as ground truth. DeepInfer inference was run in the PROMISE12 domain using the accurate ensemble mode. Although Docker GPU passthrough was available on the workstation, the model container uses an older TensorFlow runtime that did not expose a usable GPU device inside the container, so the full evaluation was completed on CPU.

At the end of the evaluation, I obtained an average whole-gland Dice score of **0.8389** across all **30** test cases. Geometry validation passed for all evaluated prediction-versus-ground-truth pairs.

Qualitative Comparison

The following screenshots are intended to visually compare the provided ground-truth annotation and the model prediction. In these screenshots, the annotation should be shown in **green**, while the model output should be shown in **red**.

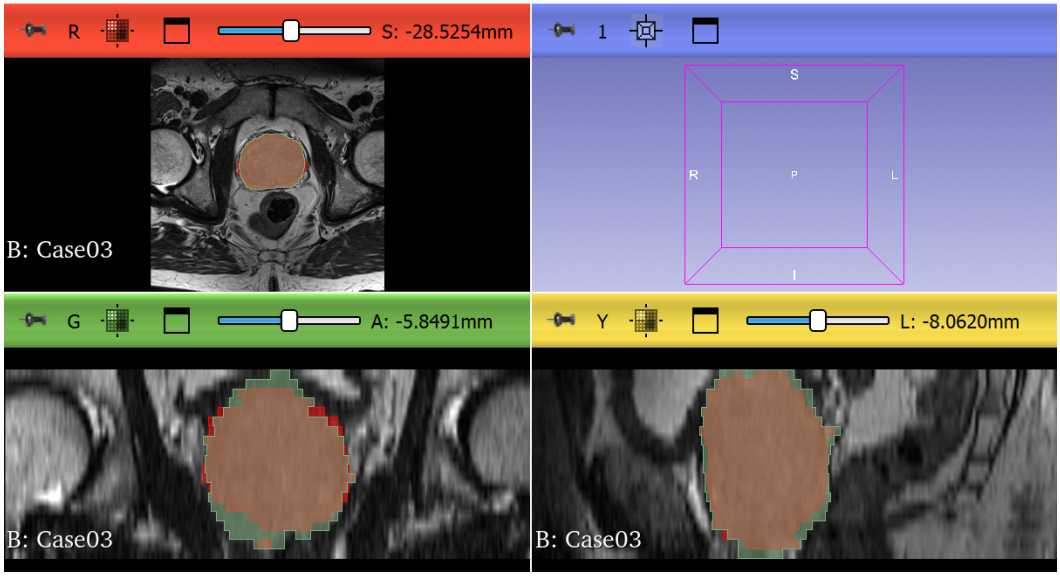

Figure 1: Comparison for Case03 between ground truth and model output. Model output in red, ground truth in green.

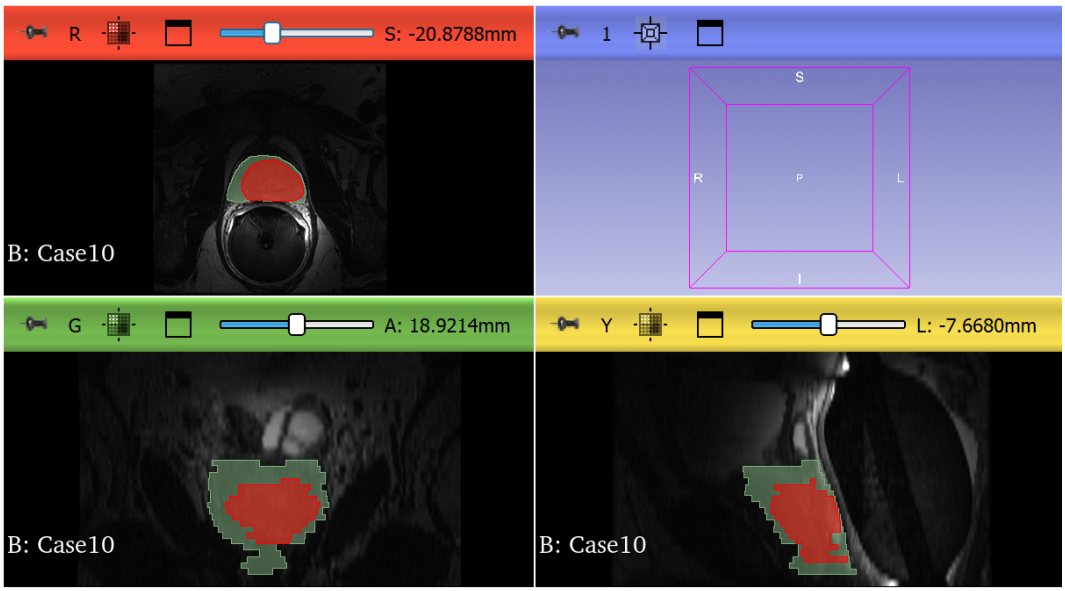

Figure 2: Comparison for Case10 between ground truth and model output. Model output in red, ground truth in green.

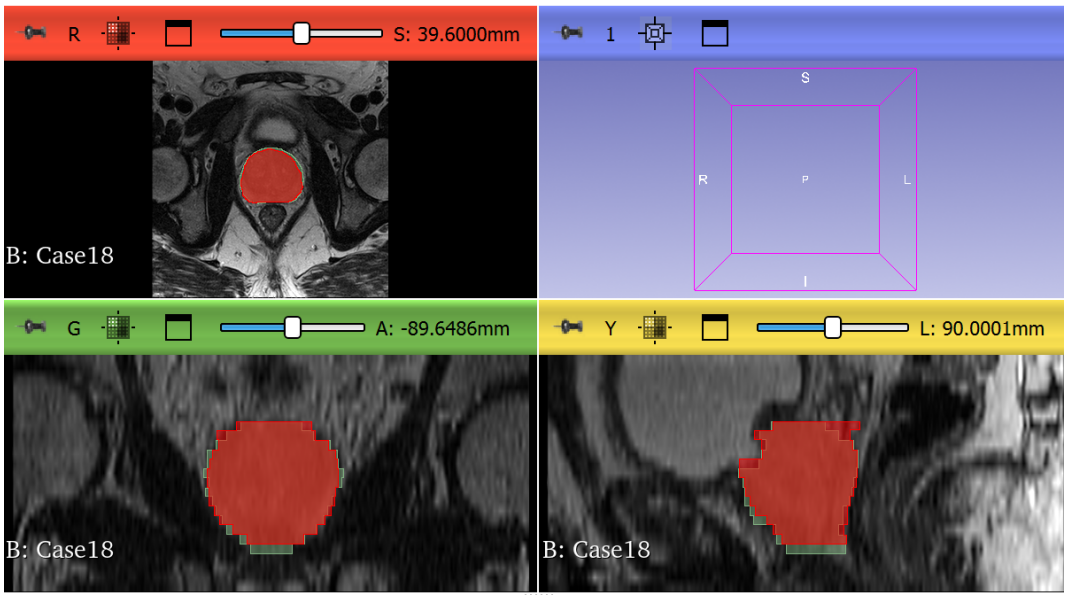

Figure 3: Comparison for Case18 between ground truth and model output. Model output in red, ground truth in green.

Quantitative Results

Table 1: Dice score results obtained on the 30-case PROMISE12 test evaluation set. Since the ground truth consists of binary whole-gland annotations, zonal Dice scores are not applicable.

Case	Mode	Ground Truth	Whole Gland Dice
`Case00`	whole-gland	binary GT	0.9338

Case	Mode	Ground Truth	Whole Gland Dice
`Case01`	whole-gland	binary GT	0.8890
`Case02`	whole-gland	binary GT	0.9158
`Case03`	whole-gland	binary GT	0.9203
`Case04`	whole-gland	binary GT	0.9241
`Case05`	whole-gland	binary GT	0.8853
`Case06`	whole-gland	binary GT	0.8962
`Case07`	whole-gland	binary GT	0.8943
`Case08`	whole-gland	binary GT	0.8699
`Case09`	whole-gland	binary GT	0.7784
`Case10`	whole-gland	binary GT	0.5210
`Case11`	whole-gland	binary GT	0.6575
`Case12`	whole-gland	binary GT	0.8787
`Case13`	whole-gland	binary GT	0.7526
`Case14`	whole-gland	binary GT	0.8013
`Case15`	whole-gland	binary GT	0.8707
`Case16`	whole-gland	binary GT	0.9139
`Case17`	whole-gland	binary GT	0.8561
`Case18`	whole-gland	binary GT	0.9150
`Case19`	whole-gland	binary GT	0.9204
`Case20`	whole-gland	binary GT	0.4413
`Case21`	whole-gland	binary GT	0.8998
`Case22`	whole-gland	binary GT	0.8729
`Case23`	whole-gland	binary GT	0.7945
`Case24`	whole-gland	binary GT	0.8215
`Case25`	whole-gland	binary GT	0.8895
`Case26`	whole-gland	binary GT	0.8544
`Case27`	whole-gland	binary GT	0.8066
`Case28`	whole-gland	binary GT	0.9245
`Case29`	whole-gland	binary GT	0.8687
Dataset mean	whole-gland	binary GT	0.8389

Applicability to Other MRI Sequences

Based on the official DeepInfer model documentation, this model is intended for **pelvic axial T2-weighted MRI scans**. The DeepInfer `Prostate-Segmenter` model page describes the task as whole-gland prostate segmentation in pelvic axial T2-W MRI scans and lists the required input as `Pelvic T2-W MRI`. In the command-line guide, the `InputVolume` is also explicitly described as a *T2-Weighted Pelvic MRI*. In addition, the official PROMISE12 dataset description states that the training cases consist of *transversal T2-weighted MR images of the prostate*. Since the PROMISE12 domain of this model was trained on that dataset, the evidence supports use on T2-weighted images rather than on other MRI sequences.

Therefore, the answer is effectively **no** for supported use on non-T2 sequences such as `ADC`, `DWI`, or `T1`-weighted MRI. The container may still accept another image file as input, but this would be an **out-of-domain use case** and there is no official evidence in the model documentation that performance on other sequences is valid or reliable. In practice, the model should be treated as a **T2-only segmentation model** unless it is retrained or separately validated on the alternative sequence.

Relevant sources: DeepInfer Prostate-Segmenter model page, PROMISE12 challenge data description.

Discussion

Overall, the model achieved strong whole-gland overlap on many of the 30 evaluated test cases, with several cases reaching Dice scores above 0.92. The best-performing case was `Case00` with a Dice score of 0.9338. The weakest result was `Case20` with a Dice score of 0.4413, and other difficult cases included `Case10` (0.5210), `Case11` (0.6575), and `Case13` (0.7526). These lower scores indicate that the model can substantially under-segment or mis-segment the gland on certain scans.

Despite these failure cases, the overall mean Dice of 0.8389 suggests that the DeepInfer model provides reasonably strong whole-gland prostate segmentation performance on this PROMISE12 test set. Since all geometry checks passed between the original `.mhd` files, the converted `.nrrd` files, and the predicted segmentations, the reported scores reflect model performance rather than file-conversion mismatch.